

ERYTHROPLASIA OF QUEYRAT

Etiology and Pathogenesis

Erythroplasia of Queyrat (EQ) is an in situ squamous cell carcinoma (SCC) that affects the mucosal surfaces of the penis in uncircumcised males. First described by Queyrat in 1911 as "erythroplasie du gland," the term *erythroplasia of Queyrat* is preferred over SCC in situ or Bowen disease (BD) due to its distinct clinical presentation—typically a well-demarcated, shiny, velvety, bright red plaque.

EQ occurs in uncircumcised men aged 20 to 80 years, with most cases occurring between the third and sixth decades of life. Risk factors include poor genital hygiene, smegma accumulation, heat, friction, trauma, and genital herpes simplex virus infection.

Human papillomavirus (HPV) plays a significant role in pathogenesis. Co-infection with HPV subtypes 8 and 16 has been identified in nearly all studied EQ lesions. Additional involvement of high-risk subtypes such as HPV-39 and/or HPV-51 has also been observed. This dual or multiple HPV infection pattern may distinguish EQ from BD and other genital neoplasias, suggesting a unique virological etiology.

AT A GLANCE

- Erythroplasia of Queyrat (EQ) is an in situ SCC of the penile mucosa in uncircumcised males.
- Clinically distinct from Bowen disease.
- Associated with HPV types 8 and 16 in most cases.
- Progresses to invasive SCC in approximately 10% of untreated cases.
- Preventive measures include circumcision and good genital hygiene.
- First-line treatment: **Mohs micrographic surgery.**

- Alternative treatments: Surgical excision, CO_2 laser vaporization, and topical 5-fluorouracil.

Clinical Findings

EQ typically presents as a glistening, red, velvety plaque on the glans penis, prepuce, or urethral meatus. About 50% of cases begin as a solitary lesion, while the rest present with multiple plaques.

Common symptoms include: - Localized pain - Pruritus - Phimosis (difficulty retracting the foreskin) - Bleeding - Crusting

These clinical features overlap with other penile conditions, necessitating biopsy for definitive diagnosis.

Histopathology

Histologically, EQ resembles Bowen disease, showing full-thickness epidermal atypia with preserved tissue architecture. However, distinguishing features may include: - Epidermal hypoplasia (thinning) - Prominent dermal plasma cell infiltrate—commonly seen in mucosal inflammatory and neoplastic conditions

These histologic nuances support the diagnosis when correlated clinically.

Diagnosis and Differential Diagnosis

The differential diagnosis includes: - Psoriasis - Seborrheic dermatitis - Lichen planus - Lichen sclerosus - Candidiasis - Plasma cell balanitis (Zoon balanitis) - Lymphogranuloma venereum - Granuloma inguinale - Syphilis - Drug eruptions

Given the broad differential, **a biopsy is essential** whenever EQ is suspected to confirm the diagnosis and rule out mimickers.

Prognosis and Clinical Course

EQ lesions tend to persist and slowly enlarge, often present for several years—average duration of 3.4 years before diagnosis in one study—before medical evaluation and biopsy.

Compared to Bowen disease, EQ has a higher risk of progression to invasive squamous cell carcinoma, estimated at **approximately 10% of cases**. Early diagnosis and appropriate treatment are crucial to prevent malignant transformation and complications.